

| Base Model Validation & Test Comparison |              |               |              |               |             |              |
|-----------------------------------------|--------------|---------------|--------------|---------------|-------------|--------------|
| Model                                   | Val_Accuracy | Test_Accuracy | Val_F1-Score | Test_F1-Score | Val_ROC-AUC | Test_ROC-AUC |
| Gradient Boosting                       | 0.936        | 0.928         | 0.789        | 0.757         | 0.973       | 0.955        |
| Random Forest (Tuned)                   | 0.906        | 0.910         | 0.662        | 0.667         | 0.937       | 0.929        |
| SVC                                     | 0.892        | 0.902         | 0.667        | 0.671         | 0.933       | 0.927        |
| Logistic Regression                     | 0.828        | 0.846         | 0.587        | 0.597         | 0.917       | 0.912        |
| KNN                                     | 0.792        | 0.796         | 0.552        | 0.536         | 0.863       | 0.858        |
| Decision Tree                           | 0.880        | 0.890         | 0.647        | 0.636         | 0.828       | 0.797        |